MODULE 2: CHAPTER 2.2 METHODS OF ERROR DETECTION (TB PG 54–62)

During data transmission across telecommunication cables or wireless links, electrical interference, attenuation, and noise can cause bits to flip ($0 leftrightarrow 1$). Five standard methods detect these errors:

2.2.1 Parity Checks & Parity Blocks (TB Pg 54–57)

- **Parity Bit:** An extra bit added to the 7-bit byte to ensure the total number of 1s is either **Even** (Even Parity) or **Odd** (Odd Parity).
- *Single Parity Limitation:* If two bits change during transmission, the parity count remains identical and the error goes completely undetected!
- **Parity Block Matrix:** Combines horizontal byte parity bits with a vertical **Parity Byte** at the bottom. The intersection of the corrupted row and corrupted column pinpoints the exact bit that flipped, enabling automatic correction without retransmission!

2.2.2 Checksum Algorithm (TB Pg 58–60)

> 1. Sender passes data blocks through an algorithm to compute a checksum value.
> 2. Checksum is appended to the packet and sent across the network.
> 3. Receiver recalculates the checksum using the same algorithm on arrival.
> 4. If the computed checksum matches the received checksum, data is error-free; otherwise, a re-transmission request is sent.

2.2.3 Echo Check (TB Pg 59–60)

The receiving computer transmits the entire received data block back to the sending computer ("echo"). The sender compares the echo with original. *Major Drawbacks:* Consumes double bandwidth, and cannot determine whether corruption occurred on the way forward or on the way back.

2.2.4 Check Digits — Modulo 11 & ISBN-13 (TB Pg 60–61)

Calculated at the end of identification numbers (barcodes, account numbers) to catch human manual data entry errors:

- **Transcription errors:** Mistyping a single digit (e.g. typing 3 instead of 8).
- **Transposition errors:** Swapping two adjacent numbers (e.g. typing 78 instead of 87).

2.2.5 Automatic Repeat Requests (ARQ) (TB Pg 61–62)

```
Protocol Flow:
1. Sender transmits packet and starts a Timeout Timer.
2. If receiver receives packet intact, it sends back a Positive
Acknowledgement (ACK).
3. If corrupted, receiver sends Negative Acknowledgement (NAK) → sender
retransmits immediately.
```